sas innovate

Microsoft Excel Proficiency with SAS®

Kristin Snyder
Sr Technical Training Consultant

#SASinnovate

Contents

Abstract **2**

Environment and Business Scenario **3**

Demo: Create a Dynamic Excel Workbook **6**

Overview *6*

Create the Data *6*

Development *6*

Production *7*

Copyright © 2026, SAS Institute Inc., Cary, North Carolina, USA. ALL RIGHTS RESERVED.

Abstract

Join us for an immersive experience where data mastery meets reporting excellence! Discover the art of seamlessly reading and writing Excel data with SAS®, unlocking a realm of possibilities. From Excel data ingestion to crafting dynamic reports in Excel using SAS®, this session empowers you to navigate the convergence of SAS® and Excel. Learn efficient SAS® techniques to harness the power of both tools, ensuring a seamless workflow for comprehensive data analysis and reporting. Elevate your skills and gain the expertise needed to unleash the full potential of SAS® in creating impactful Excel reports.

Copyright © 2026, SAS Institute Inc., Cary, North Carolina, USA. ALL RIGHTS RESERVED.

Environment and Business Scenario

Environment

This workshop is compatible with both **SAS®9** and **SAS Viya**.

Copyright © SAS Institute Inc. All rights reserved.

sas innovate 2026

Business Scenario

Process Employee HR Data

Clean, prepare, and analyze the company employee data in Microsoft Excel.

Create Monthly Excel Report

Create a final Excel report that can be accessed by the company stakeholders.

Automate the Monthly Report

Meet the final goal of automatically scheduling this report to run monthly.

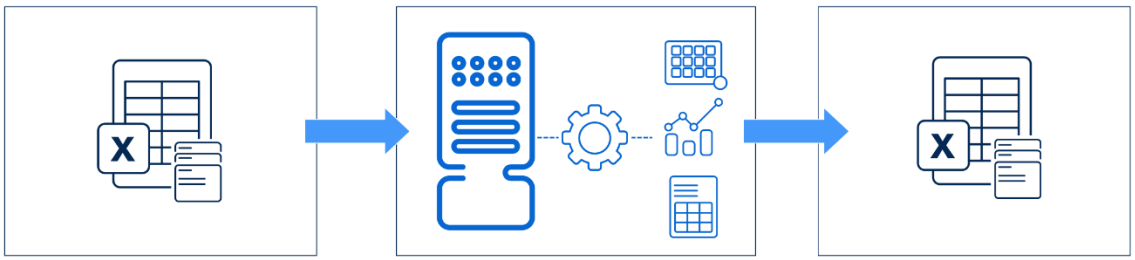
Copyright © SAS Institute Inc. All rights reserved.

sas innovate 2026

Copyright © 2026, SAS Institute Inc., Cary, North Carolina, USA. ALL RIGHTS RESERVED.

Plan

Raw Data — Data Transformations — Report



Obtain the raw Excel data from the HR system.

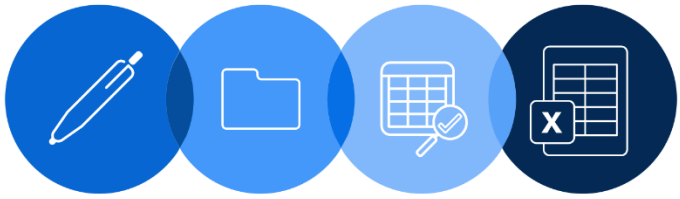
Prepare, analyze, and visualize the data using SAS.

Export the SAS results to create a final Excel report.

Copyright © SAS Institute Inc. All rights reserved.

sas innovate 2026

Project Assumptions



1 File Name
The initial Excel workbook is always named with the current year and month in which the data was extracted.

2 File Location
The raw Excel data is always exported to the same location.

3 Consistent Data
The data in each worksheet is consistent each month.

4 Program Runs Monthly
The final program will run every month and dynamically name the most recent workbook.

Copyright © SAS Institute Inc. All rights reserved.

sas innovate 2026

Copyright © 2026, SAS Institute Inc., Cary, North Carolina, USA. ALL RIGHTS RESERVED.

Demo Assumptions

Working knowledge of the **LIBNAME statement**

Ability to prepare data with the **SQL procedure**

Experience using Base **SAS procedures**

For beginners, the code in the workshop is prewritten, making it easy for you to follow along!

Copyright © SAS Institute Inc. All rights reserved.

sas innovate 2026

Copyright © 2026, SAS Institute Inc., Cary, North Carolina, USA. ALL RIGHTS RESERVED.

Demo: Create a Dynamic Excel Workbook

Overview

In this demonstration, we use **SAS Studio** to create a dynamic **Excel** workbook.

During development, we will initiate the process to generate our final workbook. To test our workflow, we'll organize each program to generate a distinct workbook for each worksheet. These workbooks will be saved within the **development\output** folder.

During production, we will consolidate all the programs that we've developed and generate a single workbook for stakeholders. Our objective is to create a single program capable of invoking each individual program and dynamically generating a comprehensive workbook with all the necessary worksheets. The resulting workbook is named based on the date of program execution and saved within the **production\output** folder.

*Note: If at any point you need to log into **SAS Viya**, use the following credentials:*

*User ID: **student***
*Password: **Metadata0***
*Assumable Groups: **No***

Create the Data

1. The machine should already be logged in to **SAS Viya** and opened to **SAS Studio**.
2. Select the **SAS Server** pane, navigate to **Files→Home→workshop→SIWEXS→data**, and double-click **cre8data.sas** to open the file.
3. On **line 13**, delete the comment and leave your cursor resting between the **equal sign (=)** and the **semi-colon (;)**.
4. In the **SAS Server** pane, right-click on the **data** folder and select **Insert as path**.
5. On the program toolbar, click **Run**.
6. On the program toolbar, click **Save**.
7. Click **X** on the program tab to close the file.
8. On the **Windows** toolbar, click the icon for the **File Explorer**. Navigate to **W:\SIWEXS\data**, double-click **2026M04_emp_info_raw.xlsx** to open the file.
   *Note: The **W:** drive is named **workshop**.*
9. Familiarize yourself with the **2026M04_emp_info_raw.xlsx** workbook. When finished, click **X** on the application to close the file.

Development

1. On the **Windows** toolbar, click the icon for **Google Chrome**.
2. In the **SAS Server** pane, navigate to **Files→Home→workshop→SIWEXS→development**, and double-click **00_config.sas** to open the file.

Copyright © 2026, SAS Institute Inc., Cary, North Carolina, USA. ALL RIGHTS RESERVED.

3. On **line 9**, delete the comment and leave your cursor resting between the **equal sign (=)** and the **semi-colon (;)**.
4. In the **SAS Server** pane, right-click on the **SIWEXS** folder and select **Insert as path**.
5. On the program toolbar, click **Run**.
6. On the program toolbar, click **Save**.
7. Click **X** on the program tab to close the file.
8. In the **SAS Server** pane, navigate to **Files→Home→workshop→SIWEXS→development**, and double-click **01_prepare_data.sas** to open the file.
   Notice on **line 21** the workbook name has been hardcoded in for the current year and month.
9. On the program toolbar, click **Run**.
10. Click the **Output Data** and **Results** tabs to familiarize yourself with the output. When finished, click **X** on the program tab to close the file.
11. Complete the following steps for the remaining 5 programs in the **development** folder.
    Note: You can ignore any warning messages that may be generated.
    a. In the **SAS Server** pane, navigate to **Files→Home→workshop→SIWEXS→development**, and double-click **##_worksheet##.sas** to open the file.
    b. On the program toolbar, click **Run**.
    c. Click **X** on the program tab to close the file.
    d. On the **Windows** toolbar, click the icon for the **File Explorer**. Navigate to **W:\SIWEXS\development\output**, double-click **worksheet_##.xlsx** to open the file.
       *Note: The **W:** drive is named **workshop**.*
    e. Familiarize yourself with the **worksheet_##.xlsx** workbook. When finished, click **X** on the application to close the file.
    f. On the **Windows** toolbar, click the icon for **Google Chrome**.

Production

1. In the **SAS Server** pane, navigate to **Files→Home→workshop→SIWEXS→production**, and double-click **create_excel_workbook.sas** to open the file.
2. On **line 37**, delete the comment and leave your cursor resting between the **equal sign (=)** and the **semi-colon (;)**.
3. In the **SAS Server** pane, right-click on the **SIWEXS** folder and select **Insert as path**.
4. On the program toolbar, click **Run**.
5. On the program toolbar, click **Save**.
6. Click **X** on the program tab to close the file.
7. On the **Windows** toolbar, click the icon for the **File Explorer**. Navigate to **W:\SIWEXS\production\output**, double-click **2026M04_HR_REPORT_FINAL.xlsx** to open the file.
   *Note: The **W:** drive is named **workshop**.*
8. Confirm all worksheets have been added to the workbook. When finished, click **X** on the application to close the file.

Copyright © 2026, SAS Institute Inc., Cary, North Carolina, USA. ALL RIGHTS RESERVED.